Section 1.3 # 35, 36

35) $v_1 = \begin{bmatrix} 20 \\ 550 \end{bmatrix}$ ← output per day of mine 1

$v_2 = \begin{bmatrix} 30 \\ 500 \end{bmatrix}$ ← output per day of mine 2

a) $5v_1$ represent the 5 day output of mine 1.

$5v_1 = 5\begin{bmatrix} 20 \\ 550 \end{bmatrix} = \begin{bmatrix} 100 \\ 2750 \end{bmatrix}$ so mine 1 outputs 100 metric tons of copper and 2750 kilograms of silver over a 5 day period.

b). $x_1 v_1 + x_2 v_2 = b$ OR $x_1 \begin{bmatrix} 20 \\ 550 \end{bmatrix} + x_2 \begin{bmatrix} 30 \\ 500 \end{bmatrix} = \begin{bmatrix} 150 \\ 2825 \end{bmatrix}$

c) $\left[\begin{array}{cc|c} 20 & 30 & 150 \\ 550 & 500 & 2850 \end{array}\right]$ $\begin{array}{l} 1/10\, R_1 \to R_1 \\ 1/10\, R_2 \to R_2 \end{array}$ $\left[\begin{array}{cc|c} 2 & 3 & 15 \\ 55 & 50 & 285 \end{array}\right]$

$1/2\, R_1 \to R_1$ $\left[\begin{array}{cc|c} 1 & 3/2 & 15/2 \\ 55 & 50 & 285 \end{array}\right]$

$-55 R_1 + R_2 \to R_2$ $\left[\begin{array}{cc|c} 1 & 3/2 & 15/2 \\ 0 & -65/2 & -255/2 \end{array}\right]$

$-2/65\, R_2 \to R_2$ $\left[\begin{array}{cc|c} 1 & 3/2 & 15/2 \\ 0 & 1 & 51/13 \end{array}\right]$

$-3/2 R_2 + R_1 \to R_1$ $\left[\begin{array}{cc|c} 1 & 0 & 21/13 \\ 0 & 1 & 51/13 \end{array}\right]$ ← I double-checked this w/ Maple. :)

36) $A = \begin{bmatrix} 27.6 \\ 3100 \\ 250 \end{bmatrix}$ $B = \begin{bmatrix} 30.2 \\ 6400 \\ 360 \end{bmatrix}$, $\begin{bmatrix} \text{Btu of heat} \\ \text{g of sulfur dioxide} \\ \text{particulate matter} \end{bmatrix}$

a) Heat: $27.6 x_1 + 30.2 x_2$

b) $x_1 A + x_2 B = x_1 \begin{bmatrix} 27.6 \\ 3100 \\ 250 \end{bmatrix} + x_2 \begin{bmatrix} 30.2 \\ 6400 \\ 360 \end{bmatrix}$

c) $\left[\begin{array}{cc|c} 27.6 & 30.2 & 162 \\ 3100 & 6400 & 23610 \\ 250 & 360 & 1623 \end{array}\right]$ ← I'm going to use maple :)

$x_1 = 3.9$ tons of A (anthracite)
$x_2 = 1.8$ tons of B (bituminous)

so: $3.9\begin{bmatrix} 27.6 \\ 3100 \\ 250 \end{bmatrix} + 1.8\begin{bmatrix} 30.2 \\ 6400 \\ 360 \end{bmatrix} = \begin{bmatrix} 162 \\ 23610 \\ 1623 \end{bmatrix}$

Section 1.4 # 42

42) To span $\mathbb{R}^4$ we need 4 pivot columns (one for each dimension). 3 vectors will give me @ most 3 pivot columns, so it will only span at most 3 dimensions. A similar argument w/ n vectors in $\mathbb{R}^m$ $(n<m)$.

PDF Problems

1.1) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \end{bmatrix}$ ← 3 vectors, all which are multiples of each other, so they span a line!

1.2) $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ ← This set may be LD, but they span $\mathbb{R}^2$ (promise).

1.3) $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ← you can remove any of the given vectors and it will span $\mathbb{R}^2$.

1.4) Not possible! Refer to exercise 42 in section 1.4 (hmm... maybe there's a reason I assigned it). We need at least 2 vectors to span all of $\mathbb{R}^2$, so removing 2 will leave us one vector, which isn't enough!

1.5) Also not possible! One of the 3 vectors will be a linear combo of the other 2. Removing any 1 vector will not be a problem, we will still span $\mathbb{R}^2$.

2) Anything goes here. I know secrets, so my answer is boring.

3) $\vec{b} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $\vec{p} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $\vec{g} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$

a) $-3\vec{b} = \begin{bmatrix} 3 \\ -6 \end{bmatrix}$ So, we take the broomstick in the opposite direction for 3 hours.

Alternatively: $-\frac{3}{2}\vec{p} + \frac{3}{2}\vec{g} = \begin{bmatrix} 3 \\ -6 \end{bmatrix}$

So, we could alternatively take the pegasus for 1.5 hours in the opposite direction, then take the maglev glider for 1.5 hours.

b) $\begin{cases} -1x + 3y + 5z = 3 \\ 2x + 4y = -6 \end{cases}$ ← this system has the infinite solution: $\begin{bmatrix} 2z-3 \\ -z \\ z \end{bmatrix}$ $(z \in \mathbb{R})$

c) I obviously choose (ii).

my explanation is that the system from (b) has infinitely many possible solutions.

For example: Take the maglev glider for one hour, pegasus one hour in the opposite direction, and the broomstick for one hour in the opposite direction.

$$-\vec{b} - \vec{p} + \vec{g} = \begin{bmatrix} 3 \\ -6 \end{bmatrix}$$ ← this uses all 3 modes of transportation!

General solution:

$$(2z-3)\vec{b} - z\vec{p} + z\vec{g} = \begin{bmatrix} 3 \\ -6 \end{bmatrix}, \quad z \in \mathbb{R}$$